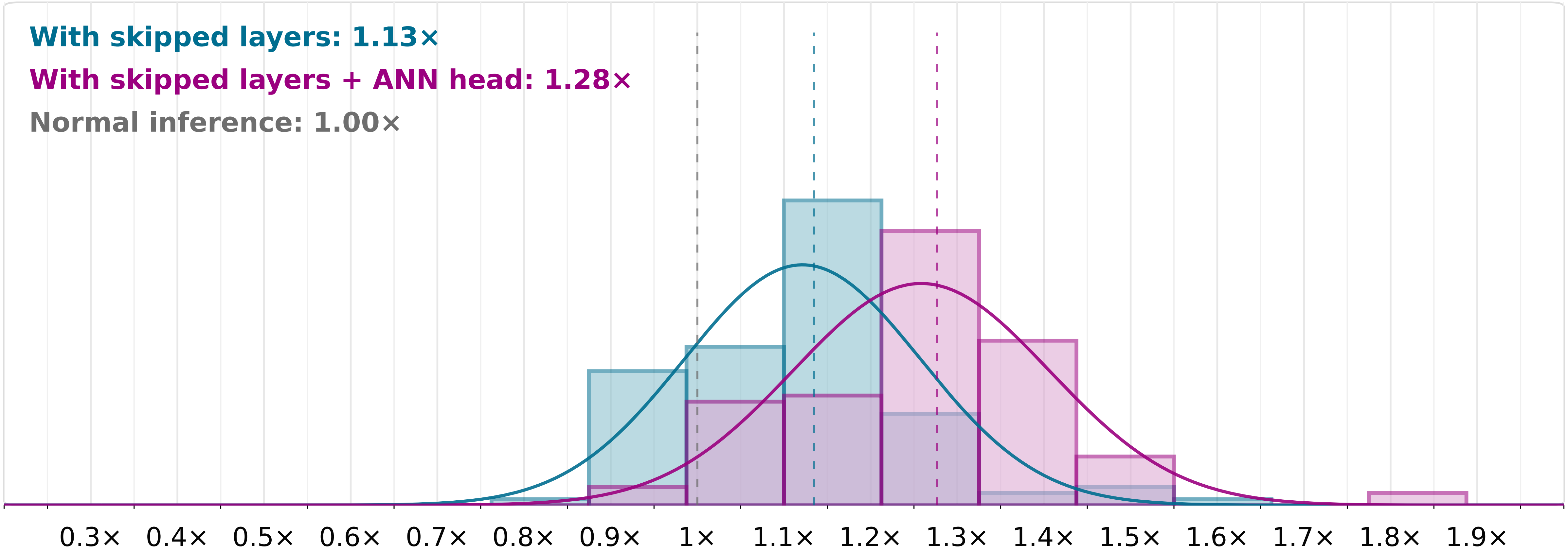
Self-speculation per-token speedup

Llama 3.2 1B Instruct



Setup

Prompt set = **concrete**
Gap = **(1, 1)**
Block size = **2**
Layers = **16**
LM vocab = **128,256**
LM params = **1.24B**
Head params = **262.7M**
Head portion = **21.25%**

Runtime

GPU = **NVIDIA L4**
Backend = **sdpa**
Model dtype = **bfloat16**
Bridge dtype = **bfloat16**
Speed internals = **no**
Profile prompts = **15**
Warmup prompts = **5**
Measured prompts = **120**

Results

Peak mem normal = **2.33 GiB**
Mean speedup (w/o ANNH) = **1.13×**
Acceptance rate (w/o ANNH) = **45.4%**
Exact match (w/o ANNH) = **58.3%**
Peak mem self (w/o ANNH) = **2.34 GiB**
Mean speedup (w ANNH) = **1.28×**
Acceptance rate (w ANNH) = **43.0%**
Exact match (w ANNH) = **58.3%**
Peak mem self (w ANNH) = **2.38 GiB**
Tokens gen = **s 17,123 / n 16,799**

Profile

Drafter split (w/o ANNH) = **B 43.9% / H 49.7% / O 6.4%**
Verifier/normal (w/o ANNH) = **1.06×**
Drafter/normal (w/o ANNH) = **33.9%**
Drafter split (w ANNH) = **B 75.1% / H 14.3% / O 10.6%**
Verifier/normal (w ANNH) = **1.06×**
Drafter/normal (w ANNH) = **19.6%**
ANNH accuracy = **94.7%**
ANNH head speedup = **5.74×**
ANNH index = **8016 clusters; top-k 100**

git commit 817a276f